

- S. Wu, J. Wu, M. Wu, K. Xiao, T. Xu, S. Yoo, K. Yu, Q. Yuan, W. Zaremba, R. Zellers, C. Zhang, M. Zhang, S. Zhao, T. Zheng, J. Zhuang, W. Zhuk, and B. Zoph. Gpt-4 technical report, 2024. URL <https://arxiv.org/abs/2303.08774>.
- J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior, 2023. URL <https://arxiv.org/abs/2304.03442>.
- C. Qian, W. Liu, H. Liu, N. Chen, Y. Dang, J. Li, C. Yang, W. Chen, Y. Su, X. Cong, J. Xu, D. Li, Z. Liu, and M. Sun. Chatdev: Communicative agents for software development, 2024. URL <https://arxiv.org/abs/2307.07924>.
- J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Y. Talebirad and A. Nadiri. Multi-agent collaboration: Harnessing the power of intelligent llm agents, 2023. URL <https://arxiv.org/abs/2306.03314>.
- X. Tang, A. Zou, Z. Zhang, Z. Li, Y. Zhao, X. Zhang, A. Cohan, and M. Gerstein. Medagents: Large language models as collaborators for zero-shot medical reasoning, 2024. URL <https://arxiv.org/abs/2311.10537>.
- H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, D. Bikel, L. Blecher, C. C. Ferrer, M. Chen, G. Cucurull, D. Esiobu, J. Fernandes, J. Fu, W. Fu, B. Fuller, C. Gao, V. Goswami, N. Goyal, A. Hartshorn, S. Hosseini, R. Hou, H. Inan, M. Kardaş, V. Kerkez, M. Khabsa, I. Kloumann, A. Korenev, P. S. Koura, M.-A. Lachaux, T. Lavril, J. Lee, D. Liskovich, Y. Lu, Y. Mao, X. Martinet, T. Mihaylov, P. Mishra, I. Molybog, Y. Nie, A. Poulton, J. Reizenstein, R. Rungta, K. Saladi, A. Schelten, R. Silva, E. M. Smith, R. Subramanian, X. E. Tan, B. Tang, R. Taylor, A. Williams, J. X. Kuan, P. Xu, Z. Yan, I. Zarov, Y. Zhang, A. Fan, M. Kambadur, S. Narang, A. Rodriguez, R. Stojnic, S. Edunov, and T. Scialom. Llama 2: Open foundation and fine-tuned chat models, 2023. URL <https://arxiv.org/abs/2307.09288>.
- K. Wang, J. Li, N. P. Bhatt, Y. Xi, Q. Liu, U. Topcu, and Z. Wang. On the planning abilities of openai’s o1 models: Feasibility, optimality, and generalizability, 2024a. URL <https://arxiv.org/abs/2409.19924>.
- M. Wang, K. Izumi, and H. Sakaji. Llmfactor: Extracting profitable factors through prompts for explainable stock movement prediction, 2024b. URL <https://arxiv.org/abs/2406.10811>.
- S. Wang, H. Yuan, L. Zhou, L. M. Ni, H.-Y. Shum, and J. Guo. Alpha-gpt: Human-ai interactive alpha mining for quantitative investment. *arXiv preprint arXiv:2308.00016*, 2023. URL <https://arxiv.org/abs/2308.00016>.
- S. Wang, H. Yuan, L. M. Ni, and J. Guo. Quantagent: Seeking holy grail in trading by self-improving large language model, 2024c. URL <https://arxiv.org/abs/2402.03755>.
- S. Wu, O. Irsoy, S. Lu, V. Dabrovolski, M. Dredze, S. Gehrmann, P. Kambadur, D. Rosenberg, and G. Mann. Bloomberggpt: A large language model for finance, 2023. URL <https://arxiv.org/abs/2303.17564>.

- Q. Xie, W. Han, X. Zhang, Y. Lai, M. Peng, A. Lopez-Lira, and J. Huang. Pixiu: A large language model, instruction data and evaluation benchmark for finance, 2023. URL <https://arxiv.org/abs/2306.05443>.
- F. Xing. Designing heterogeneous llm agents for financial sentiment analysis, 2024. URL <https://arxiv.org/abs/2401.05799>.
- A. Yang, B. Xiao, B. Wang, B. Zhang, C. Bian, C. Yin, C. Lv, D. Pan, D. Wang, D. Yan, F. Yang, F. Deng, F. Wang, F. Liu, G. Ai, G. Dong, H. Zhao, H. Xu, H. Sun, H. Zhang, H. Liu, J. Ji, J. Xie, J. Dai, K. Fang, L. Su, L. Song, L. Liu, L. Ru, L. Ma, M. Wang, M. Liu, M. Lin, N. Nie, P. Guo, R. Sun, T. Zhang, T. Li, T. Li, W. Cheng, W. Chen, X. Zeng, X. Wang, X. Chen, X. Men, X. Yu, X. Pan, Y. Shen, Y. Wang, Y. Li, Y. Jiang, Y. Gao, Y. Zhang, Z. Zhou, and Z. Wu. Baichuan 2: Open large-scale language models, 2023a. URL <https://arxiv.org/abs/2309.10305>.
- H. Yang, X.-Y. Liu, and C. D. Wang. Fingpt: Open-source financial large language models, 2023b. URL <https://arxiv.org/abs/2306.06031>.
- H. Yang, S. Yue, and Y. He. Auto-gpt for online decision making: Benchmarks and additional opinions, 2023c. URL <https://arxiv.org/abs/2306.02224>.
- H. Yang, B. Zhang, N. Wang, C. Guo, X. Zhang, L. Lin, J. Wang, T. Zhou, M. Guan, R. Zhang, and C. D. Wang. Finrobot: An open-source ai agent platform for financial applications using large language models, 2024. URL <https://arxiv.org/abs/2405.14767>.
- S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. React: Synergizing reasoning and acting in language models, 2023. URL <https://arxiv.org/abs/2210.03629>.
- Y. Yu, H. Li, Z. Chen, Y. Jiang, Y. Li, D. Zhang, R. Liu, J. W. Suchow, and K. Khashanah. Finmem: A performance-enhanced llm trading agent with layered memory and character design, 2023. URL <https://arxiv.org/abs/2311.13743>.
- Y. Yu, Z. Yao, H. Li, Z. Deng, Y. Cao, Z. Chen, J. W. Suchow, R. Liu, Z. Cui, D. Zhang, et al. Fincon: A synthesized llm multi-agent system with conceptual verbal reinforcement for enhanced financial decision making. *arXiv preprint arXiv:2407.06567*, 2024.
- B. Zhang, H. Yang, and X.-Y. Liu. Instruct-fingpt: Financial sentiment analysis by instruction tuning of general-purpose large language models, 2023a. URL <https://arxiv.org/abs/2306.12659>.
- H. Zhang, F. Hua, C. Xu, H. Kong, R. Zuo, and J. Guo. Unveiling the potential of sentiment: Can large language models predict chinese stock price movements?, 2024a. URL <https://arxiv.org/abs/2306.14222>.
- S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen, C. Dewan, M. Diab, X. Li, X. V. Lin, T. Mihaylov, M. Ott, S. Shleifer, K. Shuster, D. Simig, P. S. Koura, A. Sridhar, T. Wang, and L. Zettlemoyer. Opt: Open pre-trained transformer language models, 2022. URL <https://arxiv.org/abs/2205.01068>.
- W. Zhang, L. Zhao, H. Xia, S. Sun, J. Sun, M. Qin, X. Li, Y. Zhao, Y. Zhao, X. Cai, L. Zheng, X. Wang, and B. An. A multimodal foundation agent for financial trading: Tool-augmented, diversified, and generalist, 2024b. URL <https://arxiv.org/abs/2402.18485>.

- X. Zhang, Q. Yang, and D. Xu. Xuanyuan 2.0: A large chinese financial chat model with hundreds of billions parameters, 2023b. URL <https://arxiv.org/abs/2305.12002>.
- T. Zhong, Z. Liu, Y. Pan, Y. Zhang, Y. Zhou, S. Liang, Z. Wu, Y. Lyu, P. Shu, X. Yu, C. Cao, H. Jiang, H. Chen, Y. Li, J. Chen, H. Hu, Y. Liu, H. Zhao, S. Xu, H. Dai, L. Zhao, R. Zhang, W. Zhao, Z. Yang, J. Chen, P. Wang, W. Ruan, H. Wang, H. Zhao, J. Zhang, Y. Ren, S. Qin, T. Chen, J. Li, A. H. Zidan, A. Jahin, M. Chen, S. Xia, J. Holmes, Y. Zhuang, J. Wang, B. Xu, W. Xia, J. Yu, K. Tang, Y. Yang, B. Sun, T. Yang, G. Lu, X. Wang, L. Chai, H. Li, J. Lu, L. Sun, X. Zhang, B. Ge, X. Hu, L. Zhang, H. Zhou, L. Zhang, S. Zhang, N. Liu, B. Jiang, L. Kong, Z. Xiang, Y. Ren, J. Liu, X. Jiang, Y. Bao, W. Zhang, X. Li, G. Li, W. Liu, D. Shen, A. Sikora, X. Zhai, D. Zhu, and T. Liu. Evaluation of openai o1: Opportunities and challenges of agi, 2024. URL <https://arxiv.org/abs/2409.18486>.